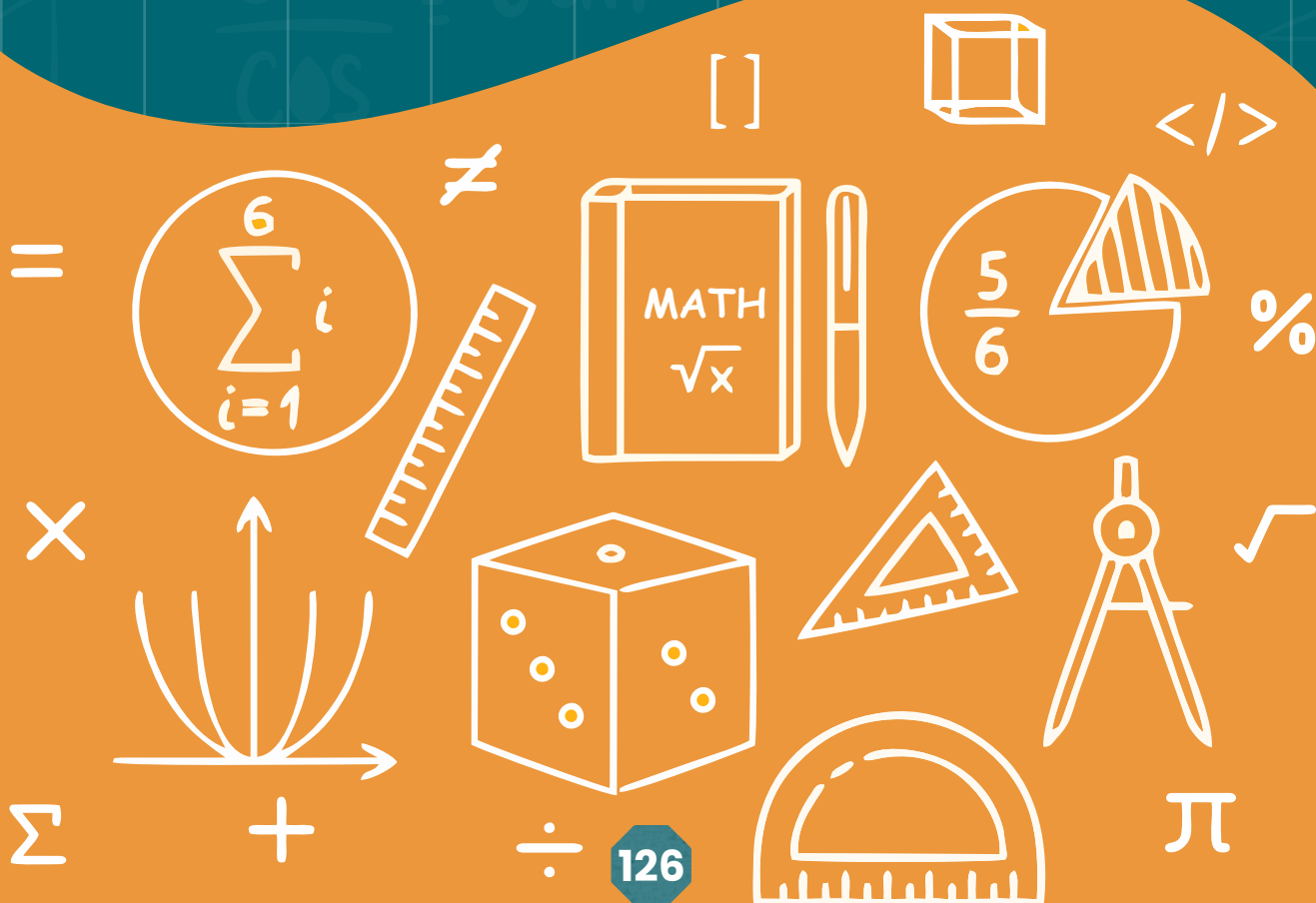


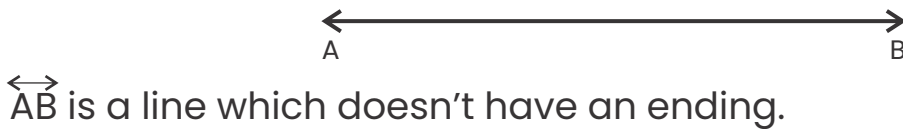
UNIT 11

PRACTICAL GEOMETRY



LINE SEGMENT

A line is a collection of points along a straight path which extends to both the directions without endpoints.



A line segment is a part of a line between two endpoints.

$\overline{PQ}$ is a line segment having P and Q as endpoints on the line AB.

Division of a line segment Definition

A line segment can be divided into 'n' equal parts, where 'n' is any natural number.

For example; a line segment of length 10 cm is divided into two equal parts by using a ruler as,

- Mark a point 5 cm away from one end.
- 10 cm is divided into two 5 cm line segments.

Similarly, a line segment of length 15 cm can be divided in the ratio 2:1 as,

- AB is the line segment of length 15 cm and C divides the line in the ratio 2:1.
- Let $CB = x$, then $AC = 2x$
- $AC + CB = 2x + x = 15$, $x = 5$
- $AC = 10$ cm and $CB = 5$ cm.
- Mark point C, 10 cm away from A

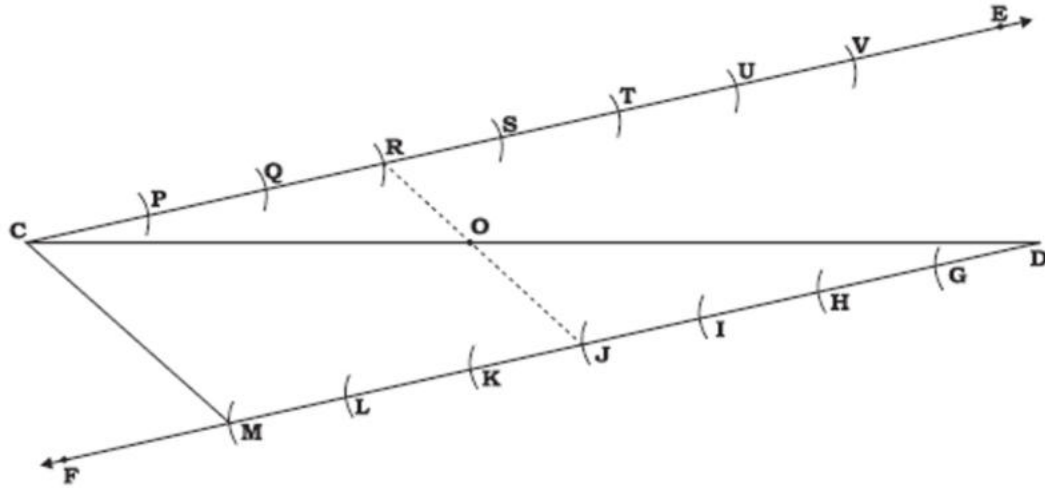
**Divide a line segment internally in a given ratio.**

Example 1: Divide a line segment CD of length 7 cm in the ratio 3:4.

Steps of construction:

1. Draw $\overline{CD}$ of length 7 cm.
2. At point C, draw $\overrightarrow{CE}$ making an acute angle $\angle DCE$ say 40° with protractor.

3. At point D, on the opposite side of $\overline{CD}$, draw $\overline{DF}$ making same acute angle $\angle CDE$ with protractor such that $m\angle CDE = m\angle CDF = 40^\circ$.



4. Using compasses, draw seven ($3 + 4 =$ the sum of ratios) arcs of suitable measure to cut $\overrightarrow{CE}$ at points P, Q, R, S, T, U, and V.
 5. Similarly, using compasses of the same measure, draw seven ($3 + 4 =$ sum of ratios) arcs to cut $\overrightarrow{DF}$ at points G, H, I, J, K, L, and M.
 6. From point C, mark the 3rd point R.
 7. From point D, mark the 4th point J.
 8. Now join points R and J and draw $\overline{RJ}$ to cut $\overline{CD}$ at point O.
 9. In this way, $\overline{CO} : \overline{OD} = 3:4$.
- It means the given $\overline{CD}$ of measure 7 cm is divided in the required ratio of 3:4 at point O or $m\overline{CO} : m\overline{OD} = 3:4$.

Example: Construct an isosceles triangle when its perimeter is 10.5 cm and the ratio of the three sides is 2:2:3.

Given: $\triangle PQR$ is an isosceles triangle. Its perimeter is 10.5 cm and the ratio among the sides is 2:2:3.

Required to construct: Draw an isosceles triangle PQR from the above data.

Steps of construction:

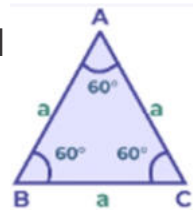
1. Draw a line segment XY measuring 10.5 cm with the help of a ruler.
2. Divide the $\overline{XY}$ in the ratio of 2:2:3. We get the measure of three sides of the triangle as $\overline{XP}$, $\overline{PQ}$, and $\overline{QY}$.

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1. Divide a line segment of length 6 cm into four equal parts.
2. Divide $m\overline{AB} = 7.5$ cm into five equal parts.
3. Divide $m\overline{CD} = 9$ cm into six equal parts.
4. Divide a line segment of 5 cm in the ratio of 2:3.

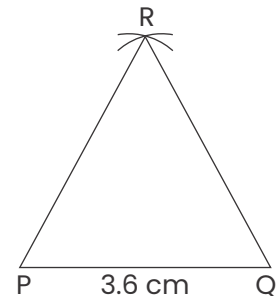
CONSTRUCTION OF EQUILATERAL TRIANGLES

An equilateral triangle is a triangle with all sides equal in length. Its three angles are also equal, and each angle measures 60 degrees.

**Construct an equilateral triangle when a side is given.**

$\triangle PQR$ is an equilateral triangle and the measure of its base is 3.6 cm.

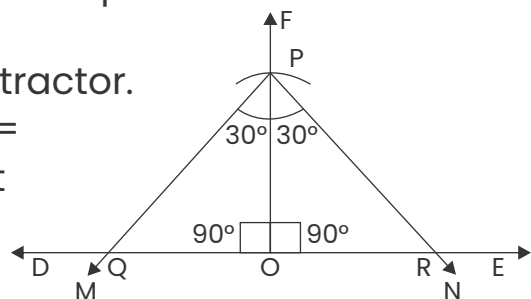
1. Draw a line segment $\overline{PQ}$ measuring 3.6 cm.
2. Take point P as centre, draw an arc with radius 3.6 cm.
3. Again take point Q as centre, draw another arc with same radius to intersect the previous arc at point R.
4. Join points P and R and draw $\overline{PR}$.
5. Join points Q and R and draw $\overline{QR}$.
6. In this way we get the required equilateral triangle PQR whose every side is of 3.6 cm.

**Construct an equilateral triangle when altitude is given.**

Example: Construct an equilateral triangle PQR when an altitude ($PO \perp QR$) where $m\overline{PO} = 4.3$ cm

Steps of construction:

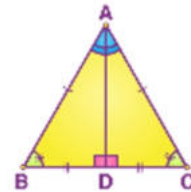
1. Draw $\overleftrightarrow{DE}$ of suitable measure with the help of a ruler.
2. Take any point O on it.
3. Draw $\overrightarrow{OF} \perp \overleftrightarrow{DE}$ with the help of a protractor.
4. Take point O as center and radius = 4.3 cm, draw an arc to cut $\overrightarrow{OF}$ at point P, to get $\overline{OP}$ as an altitude.
5. We know that every angle of an equilateral triangle is 60° and an altitude of the given equilateral triangle bisects the angle at the vertex. Draw angles of 30° with the help of the protractor at point P such that: $m\angle OPM = m\angle OPN = 30^\circ$.



6. Let $\overline{PM}$ intersect $\overline{DE}$ at point Q and $\overline{PN}$ intersect $\overline{DE}$ at point R. Thus, PQR is the required equilateral triangle.

CONSTRUCTION OF ISOSCELES TRIANGLES

An isosceles triangle is a triangle that has any two sides equal in length and angles opposite to equal sides are equal in measure.

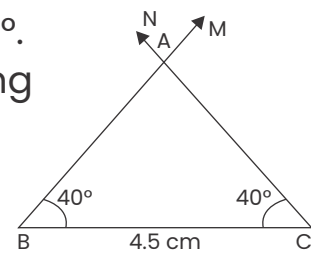


Construct an isosceles triangle when base and a base angle are given

Example: Construct an isosceles triangle ABC in which its base measures 4.5 cm and the measure of the base angle is 40° .

Steps of construction:

1. Draw $\overline{BC}$ measuring 4.5 cm.
2. At point B, draw an angle $\angle CBM$ measuring 40° .
3. At point C, also draw an angle $\angle BCN$ measuring 40° .
4. Let the arms of two angles intersect each other at point A.
5. Thus, $\triangle ABC$ is the required isosceles triangle.

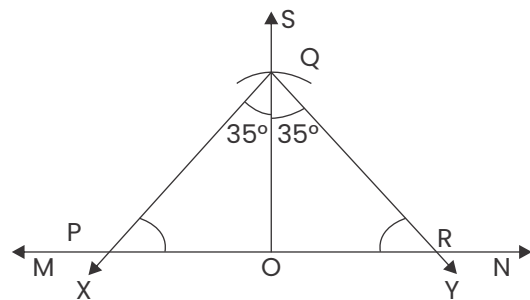


Construct an isosceles triangle when vertical angle and an altitude are given

Example: Construct an isosceles triangle PQR in which congruent sides $\overline{QP} = \overline{QR}$ (arms of an isosceles triangle) and $\overline{QO}$ is an altitude whose measure is 3.9 cm and its vertex angle measures 70° .

Steps of construction:

1. Draw $\overleftrightarrow{MN}$ with the help of a scale.
2. Take any point O on it.
3. Draw $\overline{OS} \perp \overleftrightarrow{MN}$.
4. Take point O as center, draw an arc with radius = 3.9 cm to cut $\overline{OS}$ at point Q.
5. Now as the vertex angle is 70° and altitude of the triangle



bisects the vertex angle, therefore perpendicular $\overline{OQ}$ shows the angle of $70/2 = 35^\circ$ on both sides of $\overline{OQ}$.

6. Draw $\angle OQX$ and $\angle OQY$ measuring 35° each.

7. Let $\overline{QX}$ intersect $\overline{OM}$ at point P and $\overline{QY}$ intersect $\overline{ON}$ at point R.

8. Thus, $\triangle PQR$ is the required isosceles triangle.

Construct an isosceles triangle when an altitude and a base angle are given

Example: Construct an isosceles triangle XYZ, when the measurement of its altitude XR is 3.8 cm and a base angle $\angle XYZ$ is 55° .

Steps of construction:

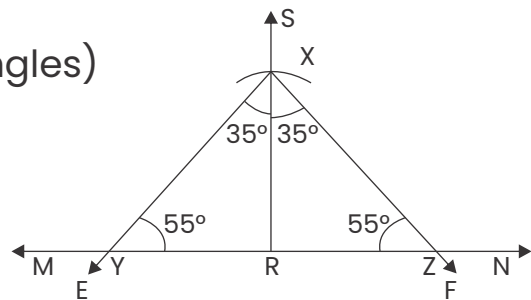
First, we find the vertex angle.

Vertex angle = $180^\circ - (\text{sum of two angles})$

Vertex angle = $180^\circ - (55^\circ + 55^\circ)$

Vertex angle = $180^\circ - 110^\circ$

Vertex angle = 70°



1. Draw $\overleftrightarrow{MN}$.

2. Take any point R on it.

3. Draw $\overline{RS} \perp \overleftrightarrow{MN}$.

4. Take point R as center and radius = 3.8 cm, draw an arc to cut $\overline{RS}$ at point X.

5. At point X, draw $\angle RXE = \text{Vertex angle}/2 = 70^\circ/2 = 35^\circ$.

6. At point X, draw $\angle RXF = \text{Vertex angle}/2 = 70^\circ/2 = 35^\circ$.

7. Let $\overline{XE}$ intersect $\overline{RM}$ at point Y.

8. Let $\overline{XF}$ intersect $\overline{RN}$ at point Z.

9. Thus, $\triangle XYZ$ is the required isosceles triangle.

EXERCISE 11.2

1. Construct the following equilateral triangles when the measure of base is as follows:

(i) 2.9 cm

(ii) 3.2 cm

(iii) 4.1 cm

2. Construct the following equilateral triangles, when the measurement of its altitude is as follows:

(i) 3.6 cm

(ii) 3.9 cm

(iii) 4.3 cm

3. Construct an equilateral triangle PQR in which the measure of its base is 50 mm long.

4. Construct an equilateral triangle MNO in which its altitude is 36 mm long.

1. Construct isosceles triangle for following data:

(i) Base = 4.2 cm, Base angle = 42°

(ii) Base = 4.8 cm, Base angle = 48°

2. Construct the following isosceles triangles when the measure of vertical angles and altitude is given:

(i) Vertical angle = 76° , Altitude = 4 cm

(ii) Vertical angle = 80° , Altitude = 5 cm

(iii) Vertical angle = 90° , Altitude = 5.3 cm

3. Construct the following isosceles triangles when the measure of altitude and base angle are given:

(i) Altitude = 4.5 cm, Base angle = 54°

(ii) Altitude = 39 mm, Base angle = 65°

PARALLELOGRAM

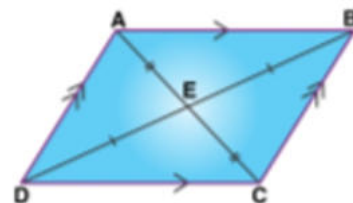
A **parallelogram** is a two-dimensional geometrical shape whose sides are parallel to each other. It is a type of polygon having four sides (also called quadrilateral), where the pair of parallel sides are equal in length. The Sum of adjacent angles of a parallelogram is equal to 180 degrees.

In the figure above, you can see, ABCD is a parallelogram, where $AB \parallel CD$ and $AD \parallel BC$.

Also, $AB = CD$ and $AD = BC$

And, $\angle A = \angle C$ & $\angle B = \angle D$

Also, $\angle A$ & $\angle D$ are supplementary angles because these interior angles lie on the same side of the transversal. In the same way, $\angle B$ & $\angle C$ are supplementary angles.



Therefore,

$$\angle A + \angle D = 180^\circ$$

$$\angle B + \angle C = 180^\circ$$

Properties of Parallelogram

- The opposite sides are parallel and equal.
- The opposite angles are equal.
- The consecutive or adjacent angles are supplementary.
- If any one of the angles is a right angle, then all the other angles will be right angles.
- The two diagonals bisect each other.
- Each diagonal bisects the parallelogram into two congruent triangles.
- The Some of the square of all the sides of a parallelogram is equal to the sum of the square of its diagonals. It is also called parallelogram law.

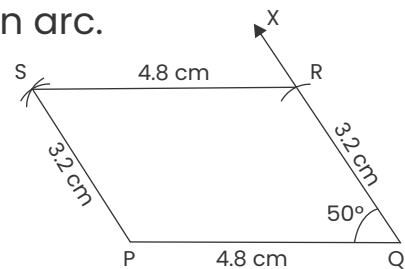
Construct a parallelogram when two adjacent sides and their included angles are given.

Example: Construct a parallelogram PQRS in which $m\overline{PQ} = 4.8$ cm, $m\overline{QR} = 3.2$ cm, and $m\angle PQR = 50^\circ$.

Given: PQRS is a parallelogram, $m\overline{PQ} = 4.8$ cm, $m\overline{QR} = 3.2$ cm, $m\angle PQR = 50^\circ$.

Steps of construction:

1. A line segment PQ measuring 4.8 cm is drawn.
2. At point Q, draw an angle $\angle PQX$ measuring 50° with a protractor.
3. With center Q and radius 3.2 cm, draw an arc to cut $\overrightarrow{QX}$ at point R.
4. With center R and radius 4.8 cm, draw an arc.
5. Again with center P and radius 3.2 cm, cut the previous arc at point S.
6. Join points P and S to draw $\overline{PS}$.
7. Join points R and S to draw $\overline{RS}$.



In this way, we get the required parallelogram PQRS.

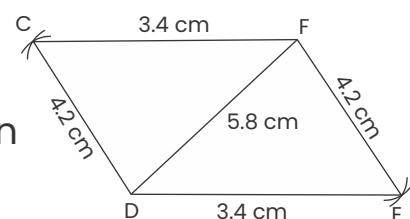
Construct a parallelogram when two adjacent sides and a diagonal are given.

Example: Construct a parallelogram CDEF in which the length of its diagonal $\overline{DF} = 5.8$ cm and the lengths of its two adjacent sides $\overline{DE} = 3.4$ cm and $\overline{EF} = 4.2$ cm.

Given: CDEF is a parallelogram. Measurement of its adjacent sides: $m\overline{DE} = 3.4$ cm, $m\overline{EF} = 4.2$ cm, and the diagonal $m\overline{DF} = 5.8$ cm.

Steps of construction:

1. Draw $\overline{DF}$ diagonal measuring 5.8 cm.
2. With centre D and radius 4.2 cm, draw an arc on the upper side of diagonal $\overline{DF}$.



3. With centre F and radius 3.4 cm, draw another arc on the upper side of $\overline{DF}$ to cut the previous arc at point C.
4. Join points D and C to draw $\overline{DC}$.
5. Join points C and F to draw $\overline{CF}$.
6. Now with centre D and radius 3.4 cm, draw an arc on the lower side of the diagonal $\overline{DF}$.
7. Again, with centre F and radius 4.2 cm, draw another arc on the lower side of the diagonal $\overline{DF}$ to cut the previous arc at point E.
8. Join points E and D to make $\overline{ED}$.
9. Join points E and F to make $\overline{EF}$.
10. In this way, we get the required parallelogram CDEF.

EXERCISE 11.3

A. Construct the following parallelograms in which the lengths of two adjacent sides and the measurement of their included angle are given as under:

1. Parallelogram PQRS, $m\overline{PQ} = 5.9$ cm, $m\overline{QR} = 4.1$ cm, $m\angle PQR = 65^\circ$.
2. Parallelogram CDEF, $m\overline{CD} = 63$ mm, $m\overline{DE} = 39$ mm, $m\angle CDE = 70^\circ$.
3. Parallelogram ABCD, $m\overline{AB} = 6.6$ cm, $m\overline{AD} = 4$ cm, $m\angle BAD = 50^\circ$.

B. Construct the following parallelograms in which the measurement of two adjacent sides and the length of one diagonal are given as under:

1. Parallelogram ABCD, $m\overline{BC} = 6$ cm, $m\overline{CD} = 4$ cm, $m\overline{BD} = 7.3$ cm.
2. Parallelogram BDFH, $m\overline{BD} = 50$ mm, $m\overline{BH} = 30$ mm, $m\overline{DH} = 64$ mm.
3. Parallelogram ACEG, $m\overline{AC} = 6.6$ m, $m\overline{AG} = 3.3$ m, $m\overline{CG} = 7.9$ m.